- Deadline: **10-October, 11:59PM**
- Submit to Moodle
- What to submit:
  - ✓ Source codes.java
  - ✓ A report containing screenshot of source code and input & output of the program

Lab 1

Getting started with Java programming

- **Instruction:** Write java programs to solve the following problems.
- **Goal:** Review programming concepts (variable, loop, condition, function, … etc.) and apply with Java programming

Lab1-Exrercise1

PRACTICE PROGRAMMING EXERCISES



Ask a user for a number, say n. Display all sequence numbers from 1000 to n. Find the sum and average of these numbers and display the result on screen. Note: Add some input validation.

Input:

Enter a number: 9

Output:

1000 999 998 997 996 995 … 11 10 9

Sum is: ….

Average is: ….



Lab1-Exrercise2

PRACTICE PROGRAMMING EXERCISES



Write a program to solve a quadratic equation $ax^2 + bx + c = 0$ using the following formula.

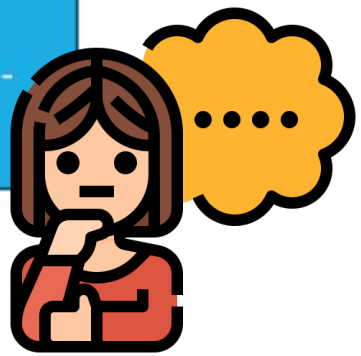
Ask the input a, b and c from the user. Then the program find roots and display them on screen.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

```
==================================
        Welcome to our program
'Quadratic Equation Solver ax² + bx + c = 0'
==================================

Input a: 2
Input b: 5
Input c: 2

----------------------------------------------------------
Roots of this equation: 2x² + 5x + 2 = 0 are:
      x1 = ?
      x2 = ?
----------------------------------------------------------
```



Lab1-Exrercise3

PRACTICE PROGRAMMING EXERCISES



Write a program that can ask users to input 5 numbers.
The program processes the input from the user and find out the minimum and maximum numbers.

Input:

7 20 100 -1 5

Output:

Among all the five input numbers (7, 20, 100, -1 and 5), min is -1 and max is 100.



Lab1-Exrercise4

PRACTICE PROGRAMMING EXERCISES



Write a program that can check if the given input date (yyyy-mm-dd) is valid or invalid.

Input: 2024-12-32
Output: invalid

Input: 2020-02-29
Output: valid

Input: 2002-14-20
Output: invalid

Input: 1991-02-28
Output: valid





Lab1-Exrercise5

PRACTICE PROGRAMMING EXERCISES



Write a program that can display all prime numbers from 2 to n, where n is an input number by user.

Input n: 10

Output:
All prime numbers from 2 to 10: 2 3 5 7

Lab1-Exrercise6

PRACTICE PROGRAMMING EXERCISES



Write a program that prompt users for inputting scores of 4 subjects for Math, English, Physics, and Programming.

a. Which courses the students got max score, min score?

b. What are the average score of this student



Input your scores of 4 subjects (Math, English, Physics, and Programming) : 90 80 65 90

Output:

Max score is: 90 (Math and Programming courses).

Min score is: 65 (Physics course).

Average score of this student is: 81.25